

Hesam Salmabadi

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[Google Scholar](#) | [ResearchGate](#) | [GitHub](#) | [LinkedIn](#)

Experience

Geospatial Data Scientist — RAP Student

May 2025 – Present

Agriculture and Agri-Food Canada (AAFC), Earth Observations, Science & Technology Branch Ottawa, ON, Canada (Remote)

- Develop machine learning models to predict agricultural land use dynamics across Canada from multi-source geospatial data
- Build reproducible Python pipelines for large-scale raster processing (xarray, dask, scikit-learn) from raw satellite ingestion to analysis
- Perform data quality control, feature engineering, and model validation on heterogeneous satellite and ancillary datasets

PhD Researcher

May 2021 – May 2026

Université du Québec à Trois-Rivières (UQTR) & Centre d'Études Nordiques, Université Laval Trois-Rivières, QC, Canada

- Developed a non-binary soil freeze–thaw detection framework using SMAP (L-band) and AMSR2 passive microwave data; validated across 87 Canadian monitoring sites spanning boreal, prairie, and tundra ecozones — match rates 78.8–98.2% (Kappa 0.44–0.87) across biome types
- Recast the Soil Freezing Characteristic Curve (SFCC) into permittivity–temperature space, enabling robust soil state classification without moisture-specific calibration
- Applied Bayesian hierarchical partial pooling and DBSCAN-based spatial block cross-validation for continental-scale model development
- Investigated 23-year dust transport climatology from the Mesopotamian Marshes using HYSPLIT forward trajectories; characterised four primary pathways with regional air quality and climate implications
- Funded by FRQNT Doctoral Research Scholarship (\$75,000 CAD, 2023–2026) and UQTR Bourse Universalis Causa

Geospatial Data Analyst — Contributing Author

Apr 2024 – Oct 2024

Environment and Climate Change Canada (ECCC) Toronto, ON, Canada (Remote)

- Contributing author, *Canada's Changing Climate Report 2026*, Chapter 6: Cryosphere Changes (forthcoming)
- Processed four decades of satellite freeze–thaw records (HDF format); performed trend analysis and produced publication-quality maps of cryosphere change across Canada

Field Research Assistant

Aug 2022

NASA Jet Propulsion Laboratory (JPL) — SMAPVEX22-Boreal Campaign Saskatchewan, Canada (On-site)

- Contributed to ground-based soil moisture and temperature measurements across the BERMS study area as part of a NASA SMAP satellite validation campaign; field data are part of Berg et al. (2025), *IEEE JSTARS*

Geospatial Data Scientist

Oct 2019 – Jan 2021

Research Institute of Cultural Heritage & Tourism (RICHT) Tehran, Iran

- Investigated long-range atmospheric transport of salt and dust particles from Lake Urmia's desiccating playa using HYSPLIT forward trajectory modelling and long-term MODIS AOD time-series analysis

GIS Data Analyst

Feb 2017 – Dec 2017

Iran University of Science and Technology (IUST), Environmental Research Laboratory Tehran, Iran

- Produced spatial distribution maps of heavy metal (Hg) contamination in Musa Estuary using ArcGIS and IDW geostatistical interpolation

Education

Doctor of Philosophy — Environmental Sciences

Université du Québec à Trois-Rivières (UQTR)

May 2021 – May 2026
Trois-Rivières, QC, Canada

Thesis: Non-Binary Monitoring of Seasonally Frozen Ground: From In Situ Sensors to Continental Passive Microwave Retrievals

Supervisor: Prof. Alexandre Roy

MSc — Civil Engineering (Environmental Engineering)

Iran University of Science and Technology (IUST)

Sep 2015 – Jul 2018
Tehran, Iran

Thesis: Determining the Effect of Wetland and Lake Desiccation in Iran on the Air Quality of Potentially Affected Regions

GPA: 18.33/20 — First Ranked Student *Supervisor:* Prof. Mohsen Saeedi

BSc — Civil Engineering

University of Birjand

Sep 2010 – Sep 2014
Birjand, Iran

Selected Publications

14 publications total (9 first-authored) | Citations: 181 | h-index: 6 | i10-index: 5 [\[Google Scholar\]](#)

Salmabadi H., Pardo Lara R., Berg A., Mavrovic A., Hanes C., Montpetit B., Roy A. (2026). In situ monitoring of seasonally frozen ground using soil freezing characteristic curve in permittivity–temperature space. *The Cryosphere*, 20, 1635–1654. [doi:10.5194/tc-20-1635-2026](#)

Salmabadi H., Berg A., Roy A. (2026, preprint). Non-Binary Retrievals of Soil Freeze–Thaw Dynamics Using Multi-Frequency Passive Microwave Data. *SSRN*. Under review. [ssrn.com/abstract=6626104](#)

Salmabadi H., Saeedi M., Notaro M., Roy A. (2025). Dust transport pathways from the Mesopotamian Marshes. *Aeolian Research*, 73, 100975. [doi:10.1016/j.aeolia.2025.100975](#)

Salmabadi H., Saeedi M., Roy A., Kaskaoutis D.G. (2023). Quantifying the contribution of Middle Eastern dust sources to PM₁₀ levels in Ahvaz, Southwest Iran. *Atmospheric Research*, 295, 106993. [doi:10.1016/j.atmosres.2023.106993](#)

Salmabadi H., Roy A., Berg A. (2026). Capturing the Soil Zero-Curtain Effect from Multi-Frequency Passive Microwave Retrievals. *ISPRS Congress 2026*.

Berg A.A., ..., **Salmabadi H.**, ... (2025). Soil Moisture Active/Passive Validation Experiment Within the Canadian Boreal Forest in 2022 (SMAPVEX22-Boreal). *IEEE J. Sel. Topics Appl. Earth Obs. Remote Sens.*, 18, 11816–. [doi:10.1109/JSTARS.2025.3564195](#)

Skills

Programming & Data: Python (xarray, dask, numpy, pandas, rasterio, geopandas, scikit-learn, matplotlib, cartopy, h5py), GDAL (Python & CLI), Bash, MATLAB, LaTeX, Git/GitHub, MySQL

Machine Learning & Statistics: Logistic regression (L1/L2), random forests, SVM, DBSCAN, Bayesian hierarchical partial pooling, probabilistic classification, Mann-Kendall trend test, spatial cross-validation

Remote Sensing: SMAP (L-band), AMSR2, MODIS AOD, Landsat time-series, OMI SO₂, ERA5, GDAS; passive microwave freeze–thaw retrieval; source apportionment (PSCF)

Modelling: HYSPLIT trajectory modelling (forward & backward); numerical atmospheric transport; soil freeze–thaw modelling

Data Formats: NetCDF, HDF4/5, GeoTIFF, Zarr, Shapefile

GIS: QGIS, ArcGIS/ArcMap, PostGIS (learning), spatial interpolation (IDW, kriging concepts)

Cloud & Infra: AWS (in progress), GitHub Actions basics, agentic AI tooling

Awards & Funding

FRQNT Doctoral Research Scholarship 2023–2026

Fonds de recherche du Québec – Nature et technologies \$75,000 CAD

Project: *Development of a New Passive Microwave Multi-Sensor Freeze/Thaw Data Product to Improve Growing Season Length Monitoring over North America*

Bourse Universalis Causa 2021–2024

Université du Québec à Trois-Rivières Up to \$21,000 CAD + tuition exemption

First Ranked Student — MSc Program 2018

Iran University of Science and Technology (IUST) GPA: 18.33/20

Full Tuition Scholarship — BSc & MSc 2010–2018

Ministry of Science, Research and Technology of Iran — awarded on national competitive examination ranking

Certificates & Professional Development

- AI for Earth Program — Amii (Alberta Machine Intelligence Institute), 2026
- Advanced Learning Algorithms — DeepLearning.AI / Coursera, 2025
- Supervised Machine Learning: Regression and Classification — DeepLearning.AI / Coursera, 2025
- Google Data Analytics Certificate (Foundations, Python, Data Exploration) — Google / Coursera, 2022–2025
- Advanced Python Programming — Maktabkhooneh, 2020

Languages

Persian (Native) | English (Fluent) | French (Functional)